Project Based Engineering Instrumentation

Using CircuitPython and the Circuit Playground

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0.1 Current Edition

This textbook was last updated on March 6, 2026. The latest pdf can be found on Github while the most up to date web based version is on Github Pages

0.2 Manuscript Changes

1. July 1st, 2020 - Original tutorials in Google Docs created
2. July 15th, 2020 - Tutorials purchased by Tangibles that Teach
3. December 21st, 2021 - Updated links for manuscript and hardware
4. May 30th, 2022 - Tangibles that teach went out of business and chapter began the move to Github
5. June 28th, 2022 - Work began on a Chromebook. Unfortunately the Figures folder is not back up on Git. As such a main_latest.pdf has been created that's the latest full version. The main.pdf is the version created by the Chromebook so it has new chapters but none of the older chapters. Figure files are now backup on Git but only figures from the 'Voltage Potentiometer' are currently there.
6. July 2nd, 2022 - All figures backed up and latest manuscript completed
7. October 18th, 2022 - A pedometer lab has been added.
8. November 8th, 2022 - Edited the Servo and feedback control servo lab to be one big lab with 3 parts
9. January 8th, 2023 - Removed all mentions of Tangibles that Teach in the main body of the text.
10. February 6th, 2023 - Changed an assignment description for LEDs and push buttons
11. February 22nd, 2023 - Moved the Bluetooth module to be before the modules lab so this will end up being a few updates to make those projects more uniform. I also created a backup called Bluetooth original in case you wanted to go back to the other version.
12. June 16th, 2023 - Cleaned up the ``changes needed" list and updated Method 3 with some new software updates.
13. August 9th, 2023 - Edited the preamble for assignments and made a note in the servo lab
14. September 5th, 2023 - Updated hyperlinks to show up on Microsoft Edge using underlines and different color text.
15. September 7th, 2023 - Fixed the TL;DR section
16. September 15th, 2023 - Updated a link in the DAQ lab
17. September 20th, 2023 - Made the modules installation section a standalone tex file
18. October 4th, 2023 - Added a new potentiometer photo
19. November 16th, 2023 - Edited the lists of parts
20. November 22nd, 2023 - Edited acceleration lab
21. December 5th, 2023 - Edited the troubleshooting section
22. January 12th, 2024 - Edited chapters 1 and 2
23. January 22nd, 2024 - Edited the bootloader update notes in the troubleshooting chatper
24. January 25th, 2024 - More edits to the bootloader notes
25. March 20th, 2024 - Added a preamble about where this textbook is located

26. April 27th, 2024 - Previously edited servo lab and pendulum lab just the assignment portion and then added this change log to the manuscript
27. May 7th, 2024 - Title page changed and many sections moved around. Added a new results and discussion section following data analysis from course surveys of this course.
28. August 12th, 2024 - Added a new requirement for the servo lab
29. September 12th, 2024 - A few edits to assignment descriptions
30. October 25th, 2024 - Updated the photocell lab to include the number of data points for the histogram
31. November 11th, 2024 - Updated Method 3 quick list
32. March 26th, 2025 - Updated servo, pitot probe and photocell assignments
33. May 24th, 2025 - Made a few changes to chapters 2 and 19 to transition between the projects and the results and discussion about lab at home frameworks and project based teaching. Also overhauled the feedback control project.
34. June 16th, 2025 - Updated the assignment directions for all projects.
35. June 17th, 2025 - Added more theory to the photocell, thermistor and pitot probe assignment
36. June 20th, 2025 - Added more figures and wiring diagrams to many assignments to help students wire up their circuits.
37. June 21st, 2025 - Added a few problems from a standard instrumentation textbook. Probably need more statistics questions
38. July 2nd, 2025 - Moved changes needed to the preamble. Added learning objectives and parts list for a few projects that had them missing.
39. September 26th, 2025 - Updated the potentiometer assignment to include converting voltage to angle in degrees.
40. October 1st, 2025 - Updated the thermistor modules assignment with the thermistor equations so it's easy to find in two places.
41. November 12th, 2025 - Fixed an equation, link and figure in the feedback control assignment
42. November 14th, 2025 - Fixed a broken link
43. November 20th, 2025 - Added a GPS requirement to the integrating acceleration lab
44. December 5th, 2025 - Added the settling time to the list of parameters for the time constant assignment and edited the overall assignment directions to have a better description of the video requirements. Also edited the servo second order systems equation for an aircraft to be stable since the equations were unstable open loop. I also moved the changes needed to Github. I also edited the purchase assignment to have bold text for some important information.
45. February 2nd, 2026 - Updated the push button and LED assignment to fix a mistake in the voltage source description.
46. February 5th, 2026 - Added an update from Jan 18th that I forgot to push which highlighted the ``Rubric" for each assignment. Also added a missing grade to the push button and LED assignment.
47. February 9th, 2026 - Moved the future project ideas to Github
48. February 26th, 2026 - Edited the trouble shooting guide for updating the CPB bootloader on Windows

0.3 Changes Needed and Future Project Ideas

Needed changes and Future Project Ideas are now tracked on Github

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Chapter 1

Introduction

This textbook describes an instrumentation kit used for an Instrumentation and Experimental Methods course at the undergraduate level. This textbook has been designed with the student and faculty member in mind. The kit contains the CircuitPlayground Express/Bluefruit running CircuitPython and is used to teach fundamentals of instrumentation and provide a hands-on way of learning. Engineering is usually taught in a traditional lecture format, involving theory in the classroom, homework outside of class, and routine examinations. Progressive forms of learning such as flipped classrooms and project based learning (PBL) have created new and fun ways for professors to interact with students and for students to be more involved in their learning. PBL provides a student-centered method of teaching and learning by posing problems for students to solve with the solution of the project being the primary goal and the theory a secondary goal.

The course begins with simple plotting and moves into data analysis, calibration and more complex instrumentation techniques such as active filtering and aliasing. This course is designed to get students away from their pen and paper and build something that blinks and moves as well as learn to process real data that they themselves acquire. There is no theory in these projects. It is all applied using the project based learning method. Students will be tasked with downloading code, building circuitry, taking data all from the ground up. By the end of this course students will be well versed in the desktop version of Python while also the variant CircuitPython designed specifically for microelectronics from Adafruit. After this course students will be able to understand Instrumentation at the fundamental level as well as generate code that can be used in future projects and research to take and analyze data. Python is such a broad and useful language that it will be very beneficial for any undergraduate student to learn this language. To the professors using this textbook, 1 credit hour labs are often hard to work into a curriculum and "live" demonstrations in the classroom cost time and money that take away from other faculty duties. I've created this kit and textbook to be completely stand-alone. Students simply need to purchase the required materials and follow along with the lessons. These lessons can be picked apart and taught sequentially or individually on a schedule suited to the learning speed of the course. The authors hope that whomever reads and learns from this textbook will walk away with an excitement to tinker, code and build future projects using microelectronics and programming. The implementation of this kit and overall teaching method has received many positive reviews from students and is reflected in anonymous course evaluations.

Students can learn in many ways with a variety of different modalities [1]. As such, instructors can choose how to present course material and have students develop content-specific skill sets. College students enter the classroom with existing skills in multi-disciplinary learning that is practiced at the secondary school level [2]. College and university professors have the opportunity to use these existing skills as a foundation for their own instructional practices. Traditional lectures provide students with lectures on theory. The instructor then provides examinations to the students based on that theory. Using this format, an instructor can focus on mathematical principles, the so-called building blocks of engineering or provide specific applications to these models [3]. In a traditional lecture format, the instructor presents theory to the students with academic problems rarely encountered in a students future career, building invaluable theoretical knowledge of concepts that appear disconnected from practical knowledge applied in the field [4]. Historically in engineering education, there has been tension between theory and application [5]. Applications include case studies or practical engineering problems that emulate future careers in engineering. This provides students with engineering phenomena that they can see and hear. They can reason through problems intuitively, memorize facts using demos, build mathematical models, or build tangible objects themselves. With these different presentation methods, students are exposed to engineering phenomena in multiple ways rather than just on the whiteboard [3]. Project Based Learning (PBL) provides an alternative for conventional teaching by

facilitating problem solving for students in a group setting that requires communication, critical thinking, and creativity. These types of problems have shown that heightened learning can happen when students interact with tangible objects that represent the theory they are presented with in the classroom [6]. Rather than learning an equation for heat transfer along a one-dimensional pipe, the students can create a pipe with thermocouples and plot data from multiple sensors along the pipe. This connection to a tangible object reinforces learning and allows students to form bridges in comprehension between their theoretical knowledge and practical application (practical knowledge) of those theories in the field [7]. Research has shown that a classroom that creates an integrated curriculum increases student satisfaction which immediately correlates to satisfactory student performance and increased graduation rates [8]. This is an example of the benefits of active student engagement versus passive student engagement. While the traditional lecture format is vital to students understandings, it is understood that application of the concepts covered in lecture are required to reach higher levels of learning [9].

Hands-on projects where students interact with tangible objects is not a new form of teaching. The laboratory environment has been around for decades. However, typically the classroom environment is separate from the laboratory environment. There has been little synergy between the two learning environments. A Mechanical Engineering degree is likely to have around three to four labs in various disciplines. In these courses, students perform an experiment in groups. They take and analyze data as well as create a report documenting their findings. Many of these laboratory experiments of course are choreographed for the student by the instructor. They follow a script and perform the experiment without building anything themselves. The students take no ownership over the experiment and there is no creativity built-in. Over the years, the nature of laboratories has changed including the lack of clear learning objectives [10]. Furthermore, this laboratory environment requires a significant amount of instrumentation and hardware to implement. For example, a shock tube or steam pump can cost tens of thousands of dollars. The maintenance and up-keep alone is not practical for smaller institutions. The teaching investment required to prepare the lab every year and the lab itself (often on the order of three hours) can be a time consuming task for a tenure track faculty member who often has a large research load.

The so-called "Lab at Home" hardware kits are becoming more and more common in the classroom to ease the burden on the instructor and institution itself [11]. These kits are small enough to be purchased and shipped to a student for them to perform experiments remotely rather than in the classroom itself [12]. They can also be brought to the classroom to be used as a personal demo aid [13]. In this sense, the take-home lab kit serves as a bridge between theory based lectures and a laboratory setting [14]. This allows both instructors and students to engage with content in a way that promotes enduring understandings and practical application of theoretical knowledge[15]. Since the kits can be used remotely, they can also be used for distance learning courses or other asynchronous activities as well as during the COVID-19 pandemic.

Two home kits in particular are directly related to instrumentation and circuits. Cyganski and Nicoletti [15] for example created a new curriculum for first year electrical engineering students by creating live demonstrations for the students. Manijikian and Simmons [13] however, combined a popular commercial microcomputer board with their own custom-designed interface board in a kit that students retain for the duration of the course for both in-lab and at-home assignments. The take-home kits however were based on the Motorola 68HC11EVB microcomputer which is programmed in assembly language. Assembly language is a rather difficult language to learn at the undergraduate level. Even with such a difficult learning curve, however, student feedback on their approach was positive [13].

It is clear that using a lab at home kit is useful for the instrumentation classroom, however the programming language to be used is a subject of debate in faculty meetings and computing committees. There are multiple languages in use today of varying complexities and use cases. There are scripting languages like Python, Ruby and MATLAB, object oriented languages like Java and C++ and compilation languages like Fortran and C. Note, this is not an exhaustive list. A recent study showed that scripting languages (Python, Ruby, MATLAB) enable writing more concise code while compilation languages (C, Fortran) create smaller executables [16]. MATLAB is often considered in engineering given its success in industry and its ability to perform numerical simulations and plotting with ease. Python, however, has become more and more popular with numerical toolboxes like numpy and plotting toolboxes like matplotlib that are free and easy to install in integrated development environments like Thonny or Spyder [17]. The Tiobe Index of Programming [18] has the top 3 programming languages listed as Python, C, and Java. MATLAB is #20 on the list. In 2004, Hans Fangohr wrote that MATLAB is much better suited than C for engineering computing but the best choice in terms of clarity and functionality is provided by Python [19]. It seems then that it would be more practical for educators even in engineering to teach the most popular language. This helps with transferability of skills and has future implications for a students' career. The scripting aspect of Python lowers complexity and allows students to learn the language quickly to apply it as a tool rather than getting stuck memorizing syntax and compilation rules. The language also comes at no cost to the students. This is a plus for students already bearing heavy financial burdens to attend a university which includes tuition and institutional

fees.

Given the success of other kits in many classes and the popularity of Python, the University of South Alabama (USA) has implemented such a kit for Instrumentation & Experimental Methods (ME 316) using CircuitPython. CircuitPython is a derivative of Python written for embedded systems and designed to simplify experimenting and learning to code [20]. The learning objectives for ME316 include: statistics, dynamic response of measurement systems, operational amplifiers, signal conditioners and fundamentals of microprocessors. This course could be taught with theory as the main focus of the lecture. However, this course includes the use of an instrumentation kit and Project Based Learning methods in addition to theory in the classroom. Each student taking the course purchases the kit and downloads a free accompanying project list [21]. Every Friday, the students complete one of the multiple projects. The following week they submit a report which includes a demo of the working project in the form of photos and videos, any plots generated if they are required to take data, all code used and a small write-up explaining their findings. The sections that follow describe the kit in more detail as well as student evaluations who have taken the course.

Note that, the Adafruit Learn page contains many tutorials on how to operate the CircuitPlayground [22]. However, Adafruit sells much more than just the CPX and thus it is often difficult for anyone to find the correct tutorial needed for the CPX. A simple search on the Adafruit Learn System yields 21 results for "servo" just on the first page with over 48 pages of results. The tutorial needed to run a servo with CircuitPython is the 10th result and it's for a different development board. Although the software works with the CPX, it does not explicitly say so. Due to the complexity of some of these systems, documentation was made and freely given to the community [21]. This documentation was custom built using a combination of multiple sources across the internet. All software for the kit is also on Github in a separate repository. Each chapter in the book contains one project for the students to complete on their own. A list of chapters and a brief description of the assignment is shown in the table of contents. Currently there are about 20 projects with multiple subsections for the students to work on during the "Funday Friday" lab days.

Chapter 2

Course Description

The course is designed to coincide with lecture content in a standard engineering instrumentation course. The students are guided through numerous projects. The specific course learning objectives, which include statistics, dynamic responses, operational amplifiers, and electronic circuits both analog and digital are shown below:

1. Apply statistical concepts of error and uncertainty analysis using normal,"t" distribution and the χ^2 distribution
2. Use propagation of error methods to determine the uncertainty of calculated quantities
3. Apply the concepts of harmonic response to predict the response of measuring systems to input signals
4. Apply the fundamentals of operational amplifiers and electronic circuits to design signal conditioning circuits including
5. Apply microprocessor fundamentals to gain an understanding of digital circuits used for digital to analog and analog to digital conversion

This course is not going to be a traditional lecture format and that hands-on projects will be assigned every Friday using the lab at home kit. The projects every Friday are created to enhance the course learning objectives as dictated above.

The course is taught on a 2+1 style where the course meets three days a week for 50 minutes each on Monday, Wednesday and Friday. On Monday and Wednesday the class is similar to what would typically be found in a standard lecture format. The instructor goes through theory and examples on a white board. On Fridays, the students come to class with their hardware kits and laptops to perform one of 20+ projects defined in the tutorials in this textbook. Fridays are nicknamed "Funday Fridays" to differentiate the format of the class and to get students excited about the projects. On Fridays, the instructor can walk around the classroom and assist students on their projects as well as highlight fundamental concepts. This is done by showing them what they learned on a whiteboard with their lab at home kit. During the COVID-19 pandemic when Universities were meeting via Zoom, class time on Friday was used to assist students on their projects while they showed their circuits via webcam in a virtual space. Since all students purchased their own kit, there was no issue with students getting hands-on experience even during COVID-19.

It is important to note here that Project Based Learning (PBL) in this course did not replace existing teaching frameworks; rather, PBL supplemented existing standard teaching measures by allowing students to apply theory to practice. The role of connectivism comes from the online networking tools that were added to the curriculum. The tutorials and assignments as well as example code are all hosted online on Github [23]. The availability of these tutorials and software opens the students to an ecosystem of free and open source software. Furthermore, the Adafruit Learning System is also used occasionally which opens them to an ecosystem of hardware and software tutorials as well as Adafruit IO (an internet of things site) and the Adafruit Forums [22]. The Adafruit Forums allow the students to comment and communicate with other groups outside the university that utilize the same hardware kits. If the instructor cannot find a solution online in a Wiki or Tutorial page, a student or the instructor can post on the forum and wait for a response from the larger Adafruit community.

Students in this course are also required to do a final project where they must create something new by using the microcontroller in the kit plus three other electrical components. One of the three components must be something not originally included in the kit. The students may also work in a group. PBL typically consists of ownership,

creativity, collaboration and critical thinking. These four aspects are clearly a big part of this lab at home kit and the final project. They can work in groups: collaboration. They must make something new: creativity. They must build the item themselves: ownership. They must apply fundamental principles of the course: critical thinking. In addition to PBL aspects, the students are exposed to the larger network of learning through Github, the Adafruit Learning system and Forums as well as general Wikis online. This network of learning is seen in the connectivism framework. The chapters that follow detail the weekly projects. The chapters that follow the weekly projects explain results from anonymous course surverys from students taking the class as well as conclusions drawn from the authors and faculty who have taught the course.

Chapter 3

Purchase Equipment

3.1 Parts List

1. Laptop with internet connectivity
2. Credit/Debit Card to Purchase items

3.2 Learning Objectives

1. Understand the world of microcontrollers
2. Learn about the Adafruit eco-system
3. Learn about the Circuit Playground Express/Bluefruit

3.3 CircuitPython Kit

In this class you're going to build some circuits that will enhance your learning experience. Rather than just solving problems by hand you're going to take and analyze data. Over the summer of 2020, I began to work with Tangibles that Teach and at the time they bundled all components together. Unfortunately, that company has gone out of business and as such for the time being you must build the kit yourself. Here is a list of all the components in the kit

1. Circuit Playground Bluefruit or Express + Included USB Cable
2. Micro Servo
3. Potentiometer
4. Photocell
5. Two Resistors (330 Ohm and 1K Ohm)
6. Alligator Clips x3
7. External Battery Pack
8. AAA Batteries x3
9. Breadboard
10. Push Button
11. LEDs x2
12. Pitot Probe

13. LSM6DS33 + LIS3MDL - 9 DoF IMU (Optional)

Below is also a photo of all the components listed above. When you get your kit familiarize yourself with all of the components. I created an unboxing video on Youtube for you to take a look. The hardware kit is a collection of multiple items. At the time of this writing the entire kit cannot be purchased however the CircuitPlayground Express (CPX) can be purchased on Adafruit [24]. A photo of the kit is shown in the figure below. The photo shows the CPX on the left inside it's protective zip lock bag.

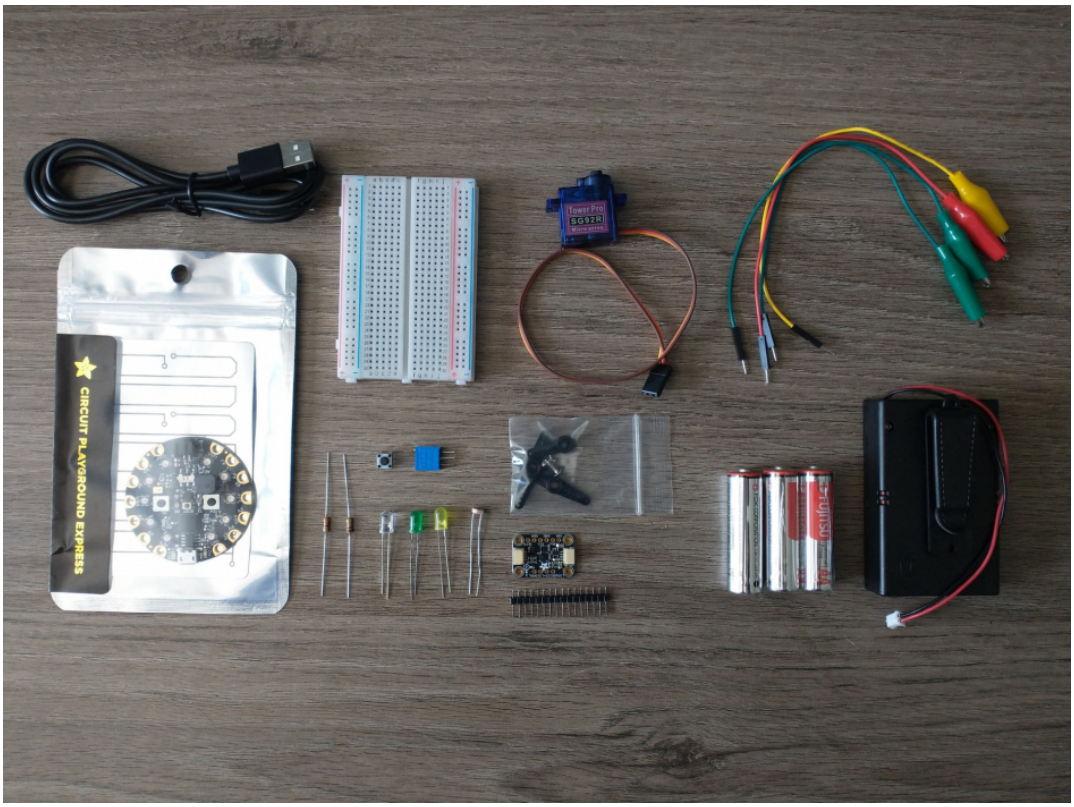

Figure 3.3.1 Photo of CircuitPython Kit Components

The CPX is a relatively new development board that operates in the same style as Arduino[25]. That is, the student can create a relatively simple script and then flash the software to the CPX via USB. The Arduino operates in the same way. The benefit of the CPX vs the Arduino is that the software is CircuitPython which is indistinguishable from Python. As mentioned earlier, Python is in the top 3 programming languages as stated by the Tiobe Index of Programming[18]. Other benefits include the components built in to the CPX. The CPX itself contains a built in microphone, accelerometer, light sensor, speaker, three push buttons, slide switch and infrared sensor. The CircuitPlayground Express (CPX) also contains 10 smart NeoPixel LEDs and a ring of copper plated pads that can be attached to the included alligator clips and breadboard. Students can also elect to purchase the CircuitPlayground Bluefruit (CPB) which does not have an infrared sensor but does contain bluetooth functionality to send data wirelessly to the Adafruit Connect App which runs on Apple/Android [26].

3.4 Assignment

Your assignment for this module is to purchase the required equipment for your class. *Note that the specific equipment will be instructor dependent so be sure you discuss with the instructor of your course before buying everything.* At a minimum you must purchase a CPX or CPB but the rest of the items in your kit will depend on which modules your instructor wants you to do during the semester. *Also, the CPX/CPB requires a microUSB cable and that cable must have a data line. Many USB cables are just power and ground.*

3.4.1 Grading Rubric

For every project you must turn in a properly formatted engineering report submitted as a PDF. The grading rubric is shown below.

1. The first page will be a title page with your name, title of the project and date - [Rubric: This is Pass/Fail]
2. The second page will contain an introduction which explains the project itself, learning objectives and expected outcomes - [Rubric: Worth 10% of your grade]
3. After the introduction you then include the project specific requirements (see below) - [Rubric: The contents below add up to 80% of your grade]
4. All figures must have appropriate figure captions, axis labels and a paragraph explaining the figure - [Rubric: Worth 10% of your grade]
5. The last few pages will be appendices which are pass/fail. Failure to include both Appendix "A" and "B" will result in a zero in the assignment. Appendix "A" will be a link (Youtube or Google Drive) to a professional video recording of you and your screen explaining the project and showing any of the systems operate as asked. *This includes any portion of the experiment such as heating up your thermistor or rotating a potentiometer and showing the response in the Serial monitor.* Appendix "B" will be any and all code used in the project - *Failure to provide code or an adequate video results in a failure of the assignment.* - [Rubric: This is Pass/Fail]

3.4.2 Specific Directions

1. A figure of your receipt of your purchases this semester (If you already own the CPX/CPB you may just take a photo of you holding the device) - [Rubric: Worth 60% of your grade]
2. Using the number of students in your class, compute the probability that at least one CPX/CPB will be shipped DOA (dead on arrival) assuming a failure rate of 1%. - [Rubric: Worth 20% of your grade]

Chapter 4

Acknowledgements

Dr. Carlos Montalvo would like to acknowledge a few key members who made this textbook possible. First and foremost I would like to thank Adafruit for their entire ecosystem of electronics, tutorials, blogs and forums. Much of what I have learned here to teach Instrumentation was from Adafruit and the Adafruit Learn system and specifically people like Lady Ada and John Park who have helped shape CircuitPython and the CircuitPlayground Express to what it is today. I would also like to thank Dr. Saami Yazdani for creating the blueprint for Instrumentation at my university by creating a laboratory environment for an otherwise totally theoretical course. His course was the foundation for this textbook and for that I thank him for showing the way. I'd like to also thank and acknowledge Tangibles that Teach for giving me the opportunity to morph this loose set of projects into a textbook that can be used for multiple universities and classrooms and of course help students learn and acquire knowledge through creating. I'd also like to change Steven Clontz for first showing me PreTeXt and for Oscar Levin and all of his helpful replies via email and Zoom. Finally, I'd like to thank the co-authors Lisa and Marine for helping me add some statistical data to this textbook and being all around great academic colleagues and friends

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